7.Create dashboards in Tableau Public Edition using the sample - Superstore dataset:

a. Sales Performance Dashboard

i. Total Sales by Region: A map showing the total sales by region.

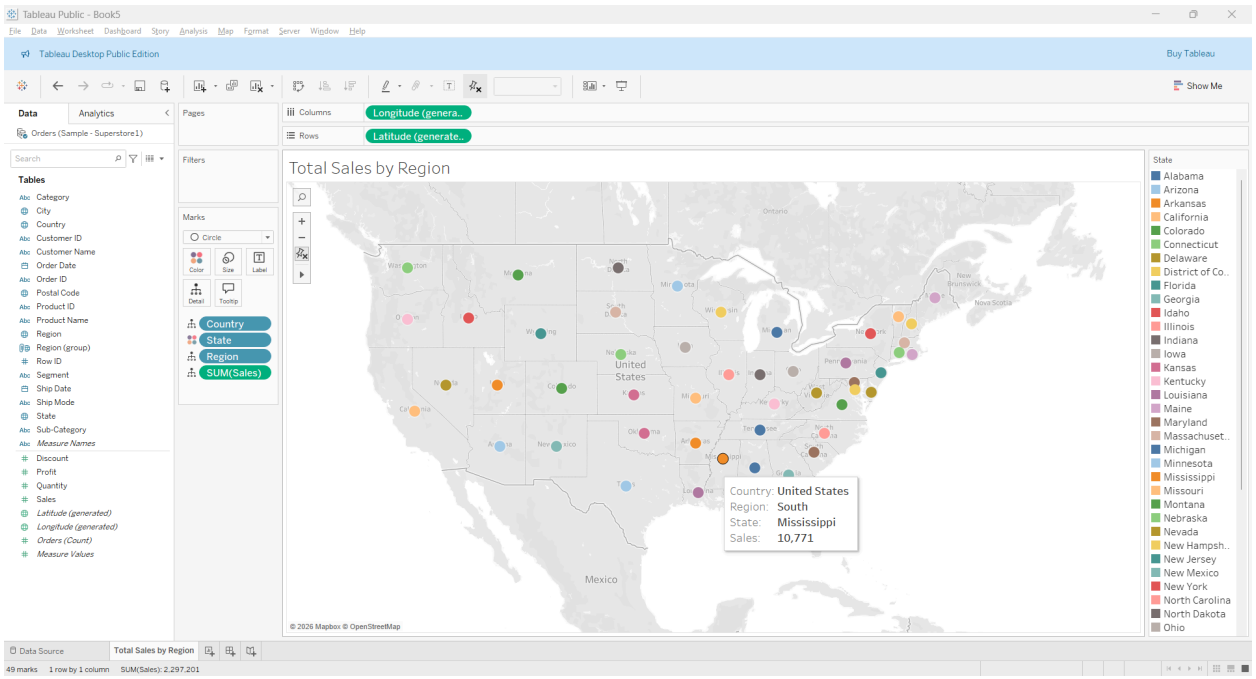
ii. Sales by Product Category: A bar chart showing the sales by product category.

iii. Top 5 Products by Sales: A table showing the top 5 products by sales.

iv. Monthly Sales Trend: A line chart showing the monthly sales trend.

Sheet 1 – Total Sales by Region (Map)

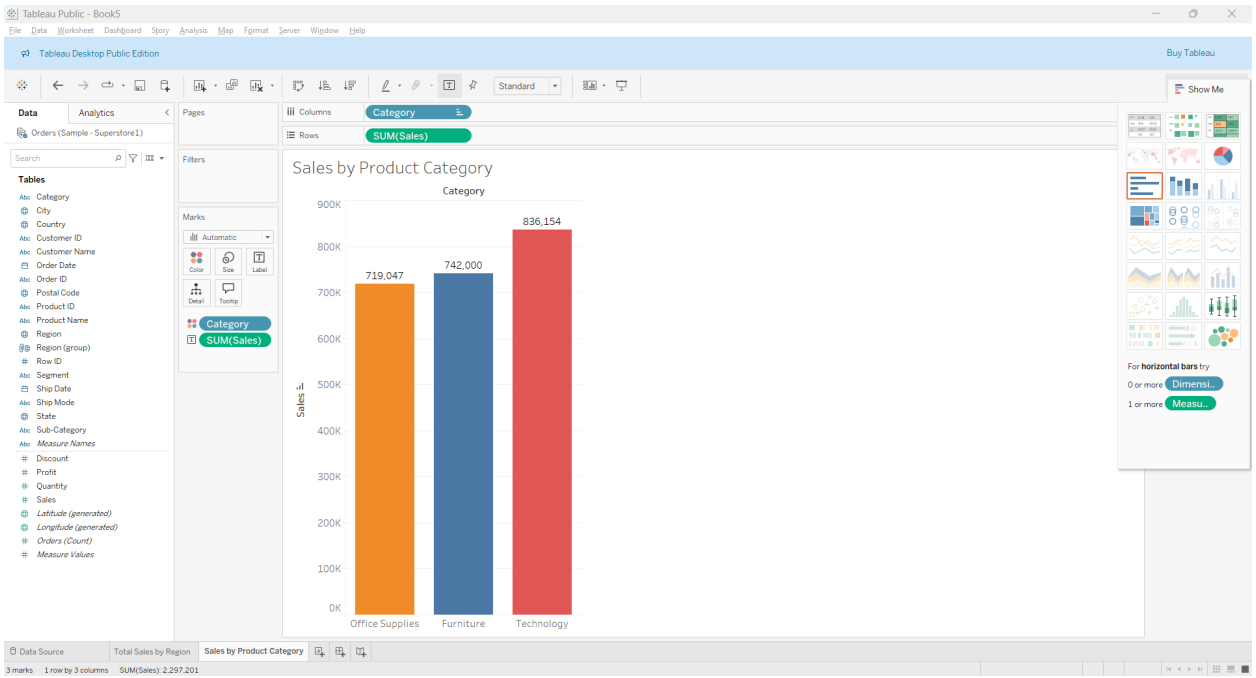
1. **Double-click Country. Tableau will generate a map.**
2. **Double-click Sales. It will added as detail in Marks**
3. **Double-click Region. It will added as detail in Marks**
4. **Drag State to Color section**
5. **Rename the sheet to Total Sales by Region.**



If need only 4 regions , as it doesn't have coordinates to show the exact location , the unknown region will be shown in selected place **: Refer : Question D) 2nd Sheet**

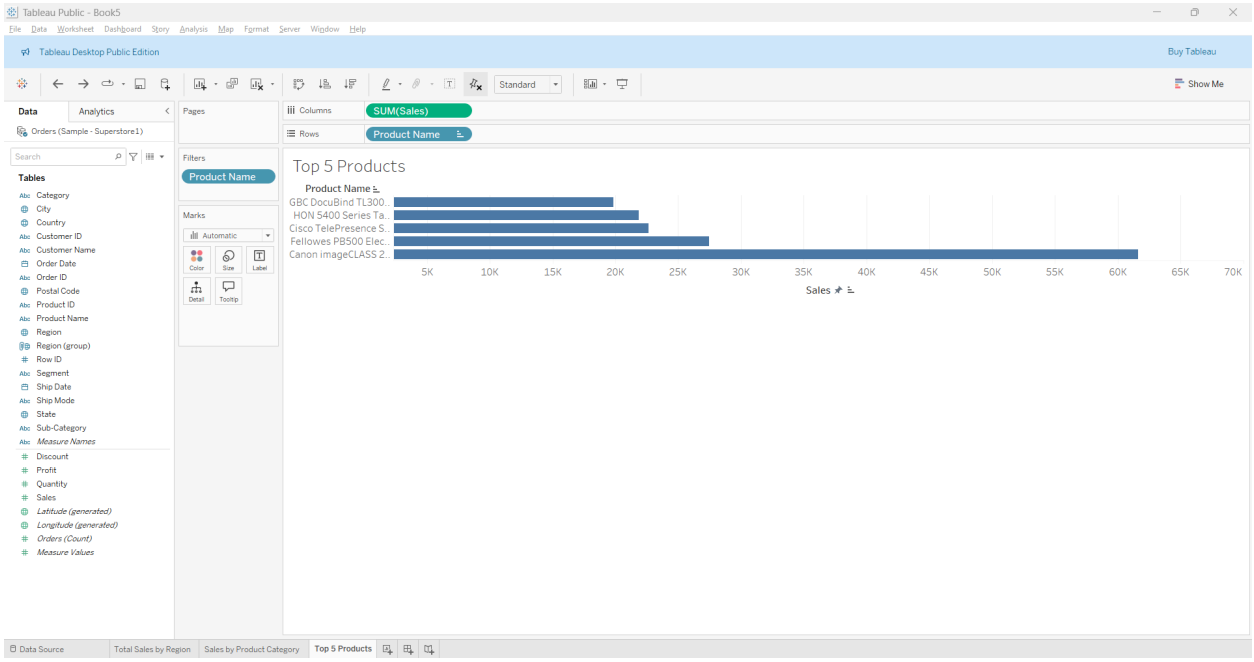
Sheet 2 – Sales by Product Category (Bar Chart)

1. **Click the New Worksheet icon at the bottom.**
2. **Drag Category to the Columns shelf.**
3. **Drag Sales to the Rows shelf.**
4. **Click the Sort Ascending icon in the top toolbar to rank the categories.**
5. **Drag Sales to the Label card to show the exact numbers on the bars.**
6. **Drag Category to the Colors Mark.**
7. **Rename the sheet to `Sales by Product Category`.**



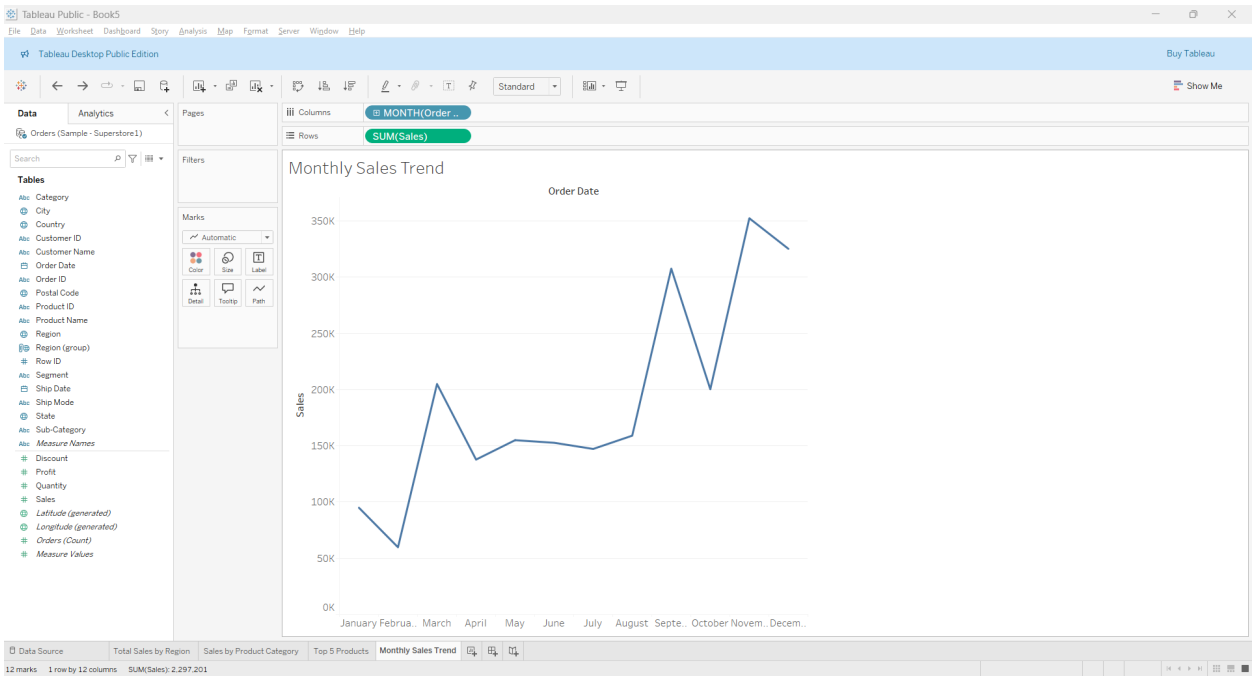
Sheet 3 – Top 5 Products by Sales (Table)

1. Click the New Worksheet icon.
2. Drag Product Name to the Rows shelf.
3. Drag Sales to the Column Shelf.
4. To Filter Top 5:
    - Drag Product Name into the Filters shelf.
    - Select the Top tab.
    - Select By field→ Top → type 5.
    - Ensure it is set to Sales and Sum.
5. Click OK.
6. Rename the sheet to Top 5 Products.



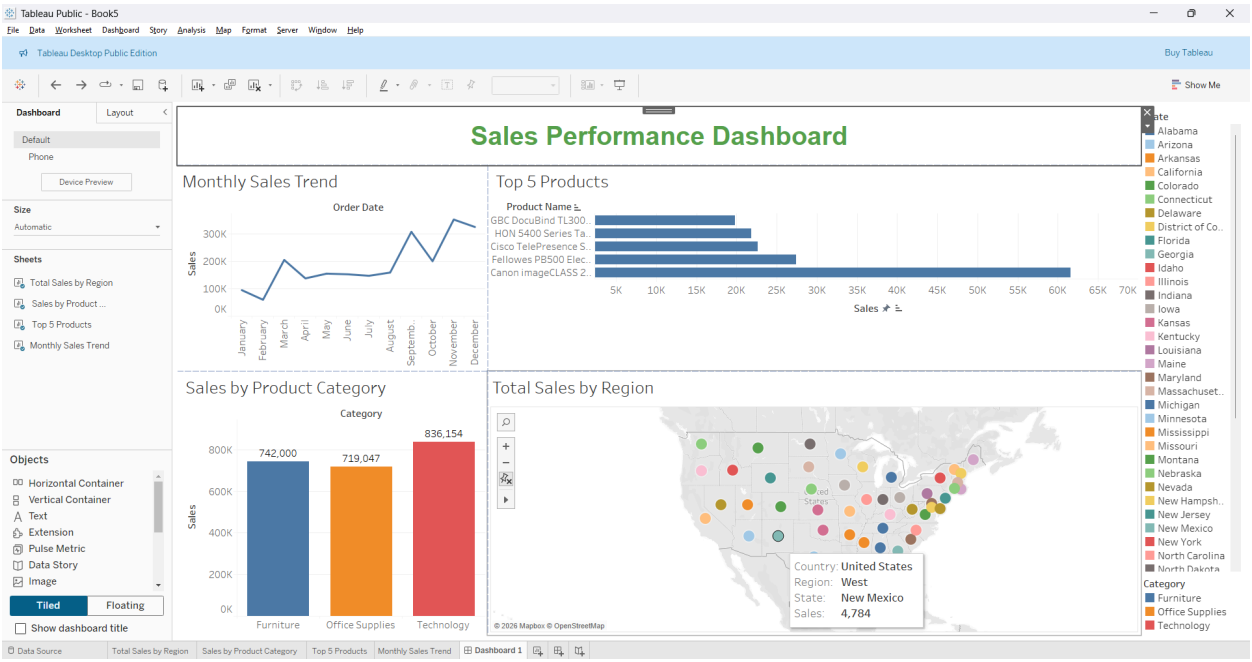
Sheet 4 – Monthly Sales Trend (Line Chart)

1. **Click the New Worksheet icon.**
2. **Drag Order Date to the Columns shelf.**
   - ***Tip:*** **Click the small arrow on the YEAR(Order Date) pill and select the second Month option to create a continuous line.**
3. **Drag Sales to the Rows shelf.**
4. **Change the Mark type to Line if it isn't already.**
5. **Rename the sheet to `Monthly Sales Trend`.**



Create the Dashboard

1. **Click the New Dashboard icon (next to New Worksheet).**
2. **Set Size to "Automatic" in the Dashboard pane so it fits your screen.**
3. **Drag and Drop:**
    - **Drag the Map to the top left.**
    - **Drag the Bar Chart to the top right.**
    - **Drag the Line Chart to the bottom left.**
    - **Drag the Top 5 Table to the bottom right.**
4. **Add a Title: Drag a Text object to the very top and type "Sales Performance Dashboard".**
5. **Enable Interactivity (The Funnel Icon):**
    - **Click on the Sales by category container in the dashboard.**
    - **Click the small Funnel icon ("Use as Filter") on the top right of the map border.**

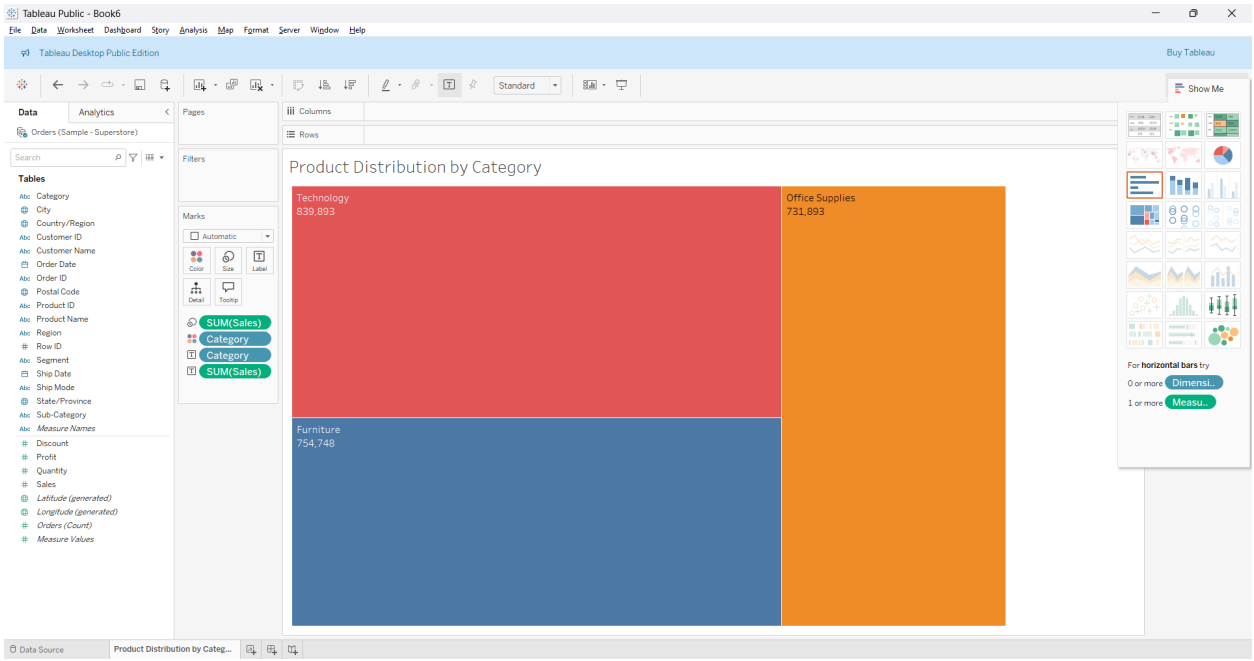


b. Product Analysis Dashboard
i. Product Distribution by Category: A Treemap showing the product distribution by category.
ii. Product Distribution by Sub-Category: A Treemap showing the product distribution by sub-category.
iii. Top 5 Products by Profit: A table showing the top 5 products by profit.
iv. Product Sales by Quarter: A bar chart showing the product sales by quarter.

Sheet 1 – Product Distribution by Category (Treemap)

A Treemap is excellent for showing how much each category contributes to the whole.

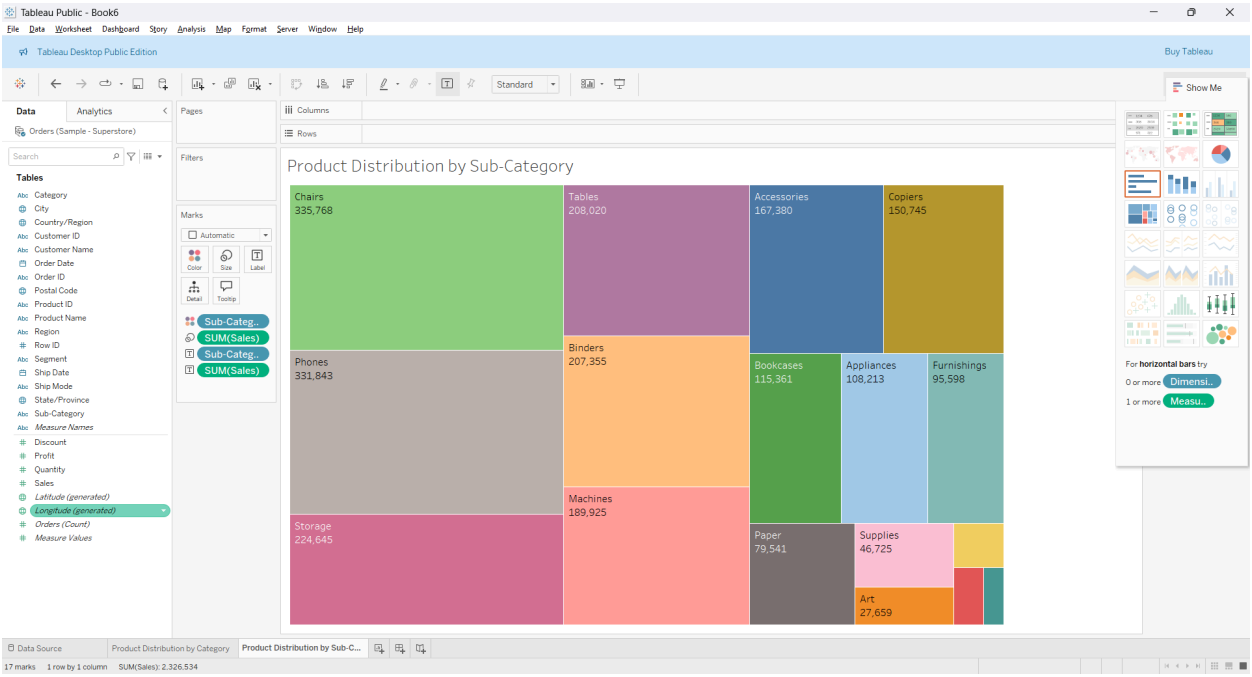
1. Open a **New Worksheet**.
2. Drag **Category** to the **Label** card.
3. Drag **Category** to the **Color** card.
4. Drag **Sales** to the **Label** card.
5. Drag **Sales** (or *Quantity*) to the **Size** card.
6. If it doesn't automatically look like a Treemap, go to the **Show Me** panel (top right) and select the **Treemap** icon.
7. Rename the sheet: `Product Distribution by Category`.



Sheet 2 – Product Distribution by Sub-Category (Treemap)

This provides a deeper look at the products.

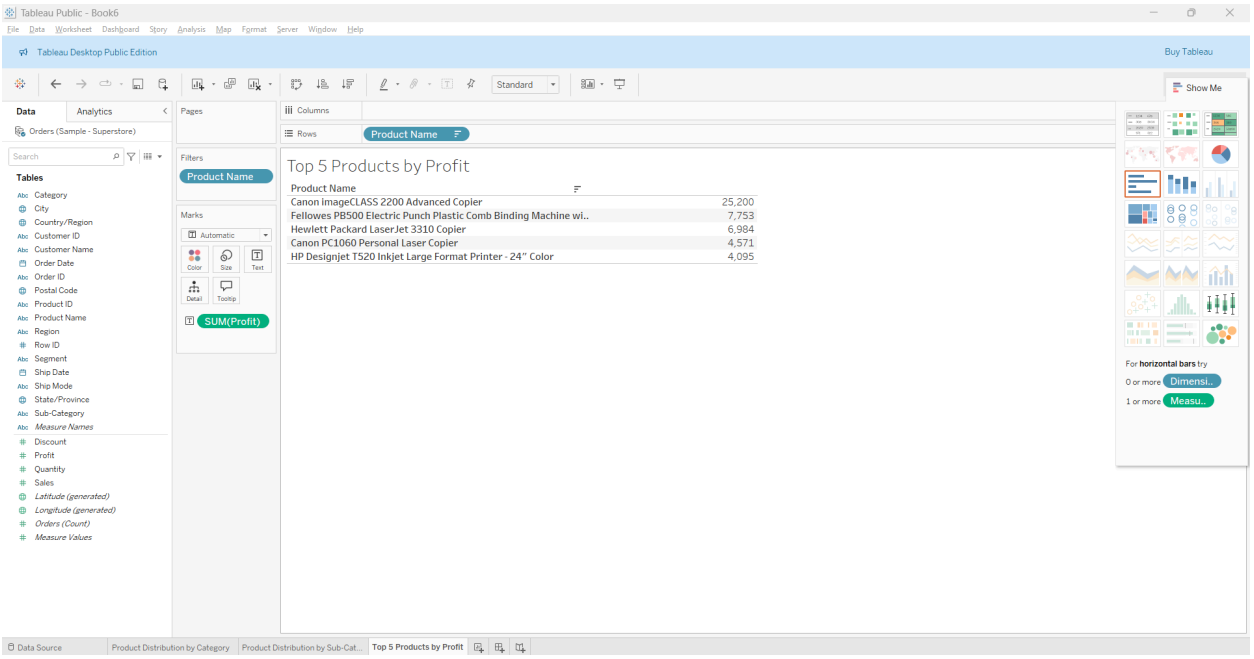
1. Open a **New Worksheet**.
2. Drag **Sub-Category** to the **Label** card.
3. Drag **Sales** to the **Size** card.
4. Drag **Sales** to the **Label** card.
5. Drag **Sub-Category** to the **Color** card.
    - *Pro-Tip:* Using Category for color here helps you see which sub-categories belong to the same parent group (e.g., all Furniture sub-categories will be the same color).
6. If needed, select **Treemap** from the **Show Me** panel.
7. Rename the sheet: `Product Distribution by Sub-Category`.



Sheet 3 – Top 5 Products by Profit (Table)

This identifies your most successful individual items.

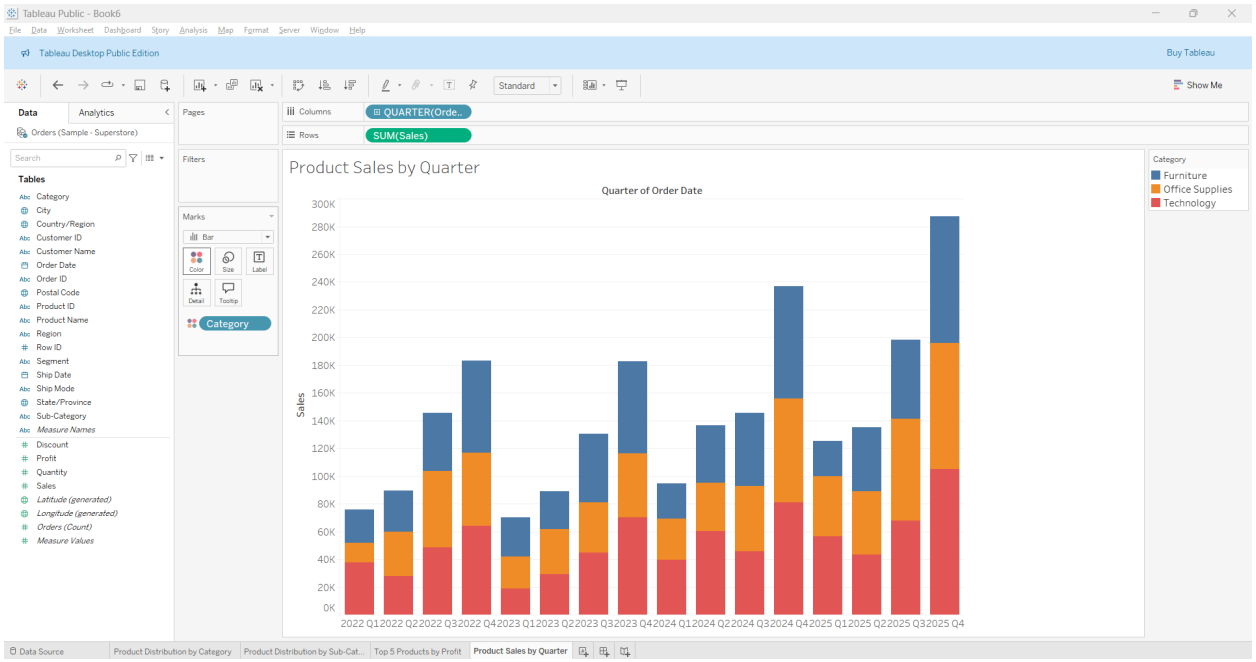
1. Open a **New Worksheet**.
2. Drag **Product Name** to the **Rows** shelf.
3. Drag **Profit** to the **Text** card.
4. **Filter for Top 5:**
    - Drag **Product Name** into the **Filters** shelf.
    - Go to the **Top** tab.
    - Select **By field** --> **Top** --> 5.
    - Ensure it is set to **Profit** and **Sum**.
5. Click **OK**.
6. Sort the table by clicking the sort icon on the **Profit** column.
7. Rename the sheet: Top 5 Products by Profit.



Sheet 4 – Product Sales by Quarter (Bar Chart)

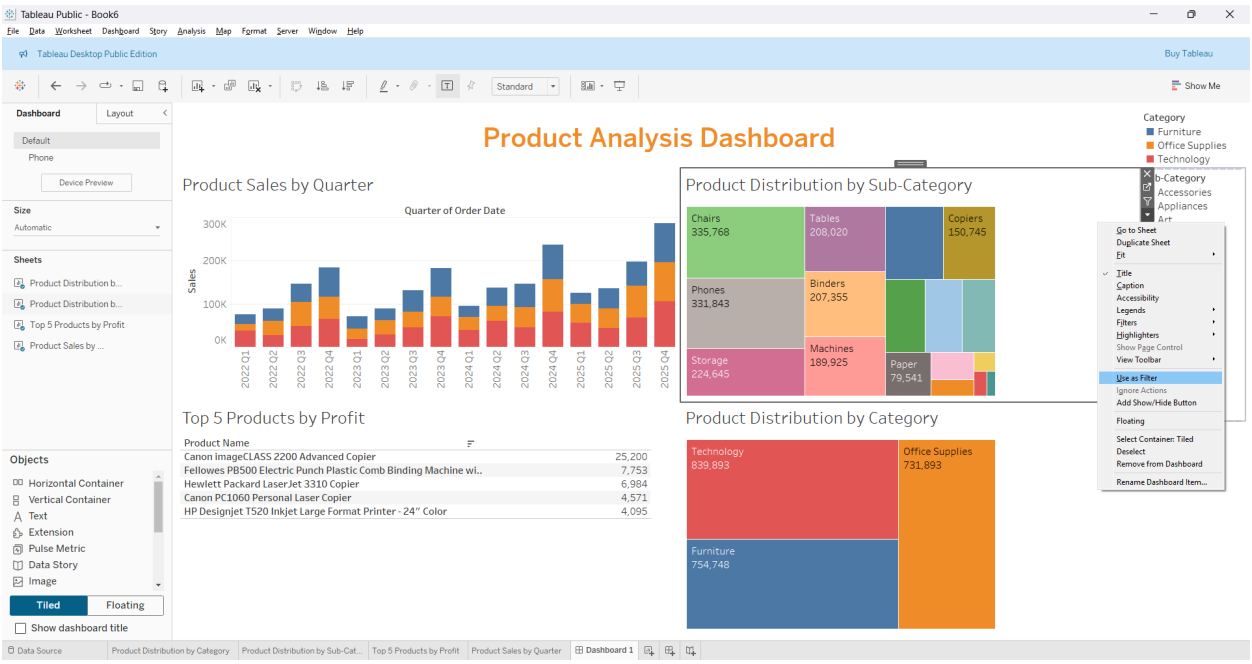
This shows seasonal trends in your product sales.

1. Open a **New Worksheet**.
2. Drag **Order Date** to the **Columns** shelf.
    - By default, it shows "Year". Click the + sign on the Year pill to expand it to **Quarter**.
3. Drag **Sales** to the **Rows** shelf.
4. Change the Mark type to **Bar** (Tableau might default to a Line chart for dates).
5. Drag **Category** to **Color** to see how different categories perform each quarter.
6. Rename the sheet: `Product Sales by Quarter`.



Assemble the Product Analysis Dashboard

1. Click the **New Dashboard** icon.
2. Set the **Size** to **Automatic**.
3. **Layout:**
    - Place the two **Treemaps** side-by-side at the top.
    - Place the **Quarterly Bar Chart** in the middle.
    - Place the **Top 5 Table** at the bottom.
4. **Add Interactivity:**
    - Select the **Sub-Category Treemap** and click the **Use as Filter** (funnel) icon.



c. Time-Based Analysis Dashboard

i. Monthly Sales Trend: A line chart showing the monthly sales trend.

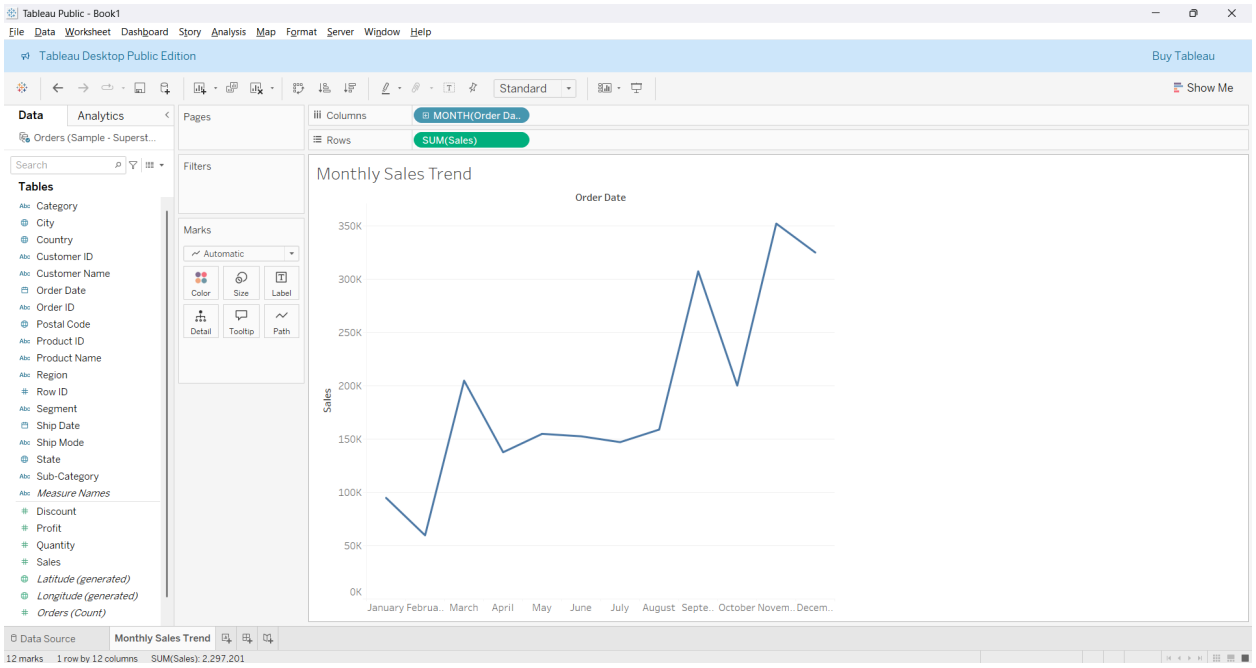
ii. Quarterly Sales Trend: A line chart showing the quarterly sales trend.

iii. Yearly Sales Trend: A line chart showing the yearly sales trend.

iv. Top 5 Quarters by Sales: A table showing the top 5 quarters by sales.

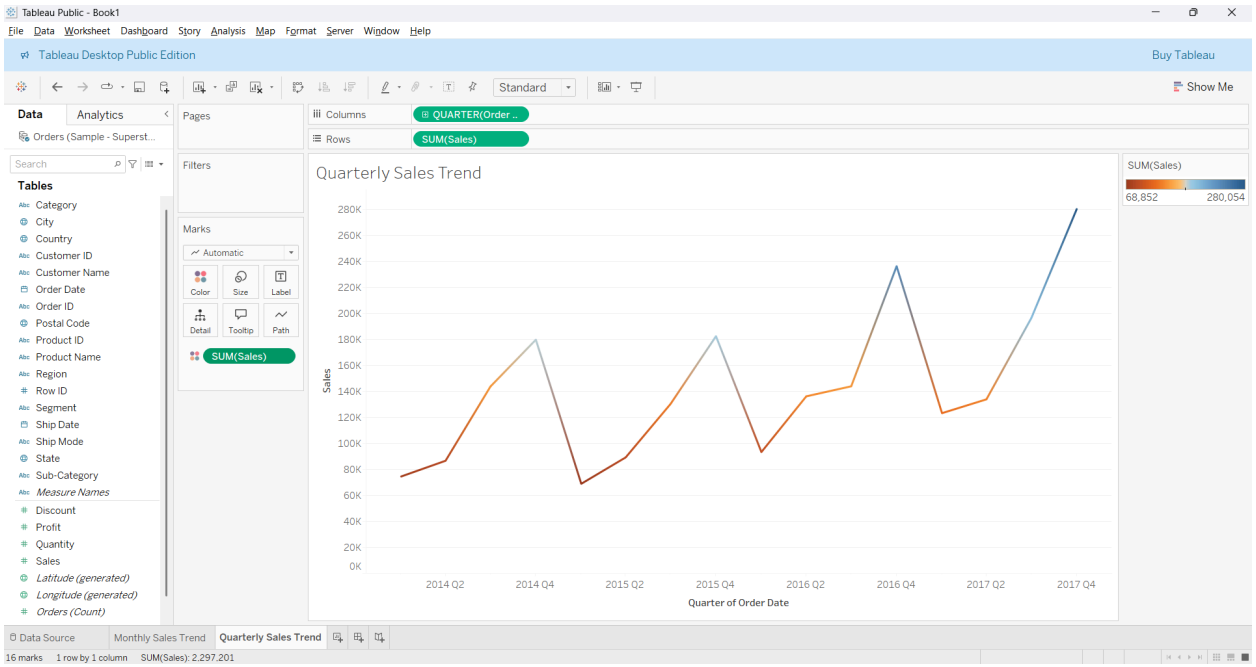
Sheet 1 – Monthly Sales Trend (Line Chart)

1. Click the **New Worksheet** icon.
2. Drag **Order Date** to the **Columns** shelf.
3. Click the small arrow on the Date pill and select the **second "Month"** option (the one that says "May 2015" in the preview). This creates a "Continuous" date that shows the trend over the full timeline.
4. Drag **Sales** to the **Rows** shelf.
5. Rename the sheet: `Monthly Sales Trend`.



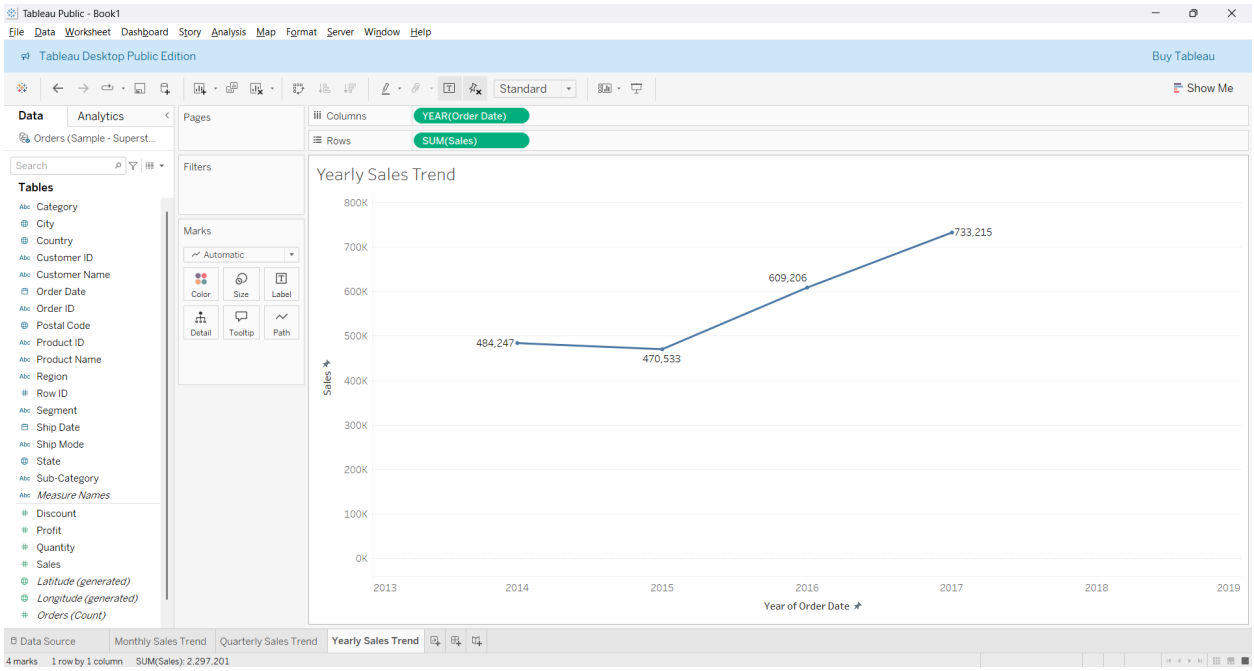
Sheet 2 – Quarterly Sales Trend (Line Chart)

1. Click the **New Worksheet** icon.
2. Drag **Order Date** to the **Columns** shelf.
3. Click the small arrow on the Date pill and select the **second "Quarter"** option (Continuous).
4. Drag **Sales** to the **Rows** shelf.
5. **Color Tip:** Drag **Sales** to the **Color** card to add a visual gradient to your line.
6. Rename the sheet: `Quarterly Sales Trend`.



Sheet 3 – Yearly Sales Trend (Line Chart)

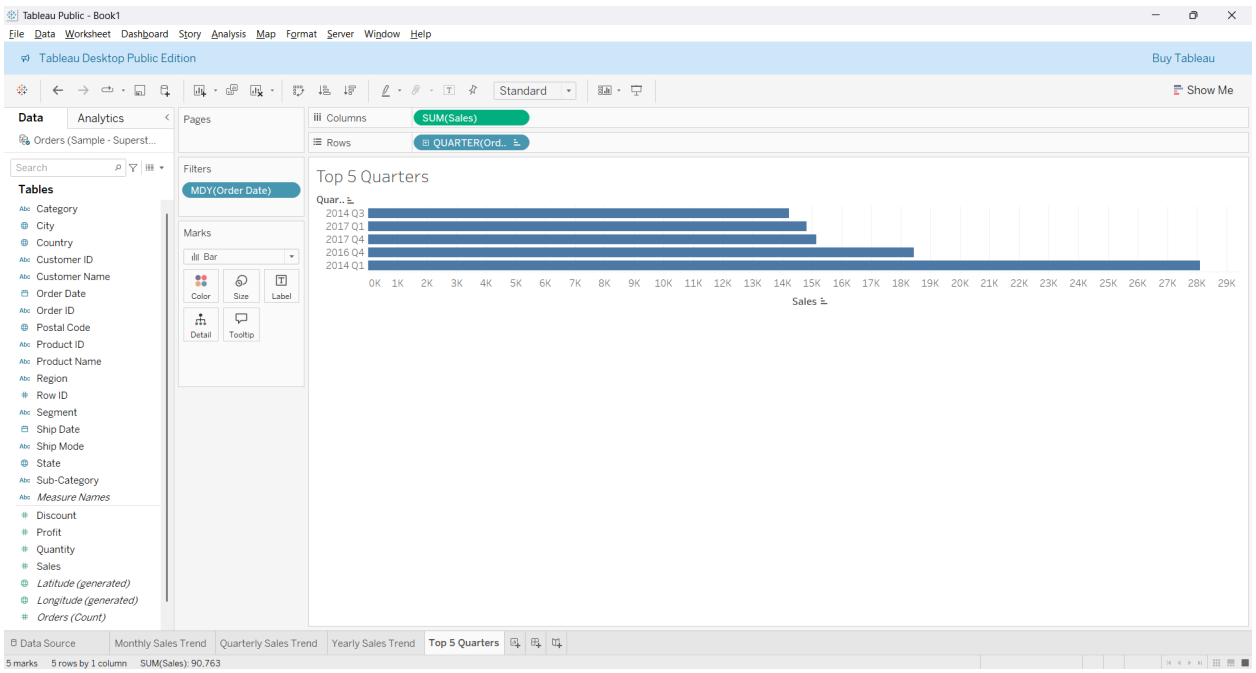
1. Click the **New Worksheet** icon.
2. Drag **Order Date** to the **Columns** shelf.
3. Keep it as **Year** (Discrete or Continuous).
4. Drag **Sales** to the **Rows** shelf.
5. Click the **Label** card and check **Show mark labels** so the total for each year is visible.
6. Rename the sheet: `Yearly Sales Trend`.



Sheet 4 – Top 5 Quarters by Sales (Table)

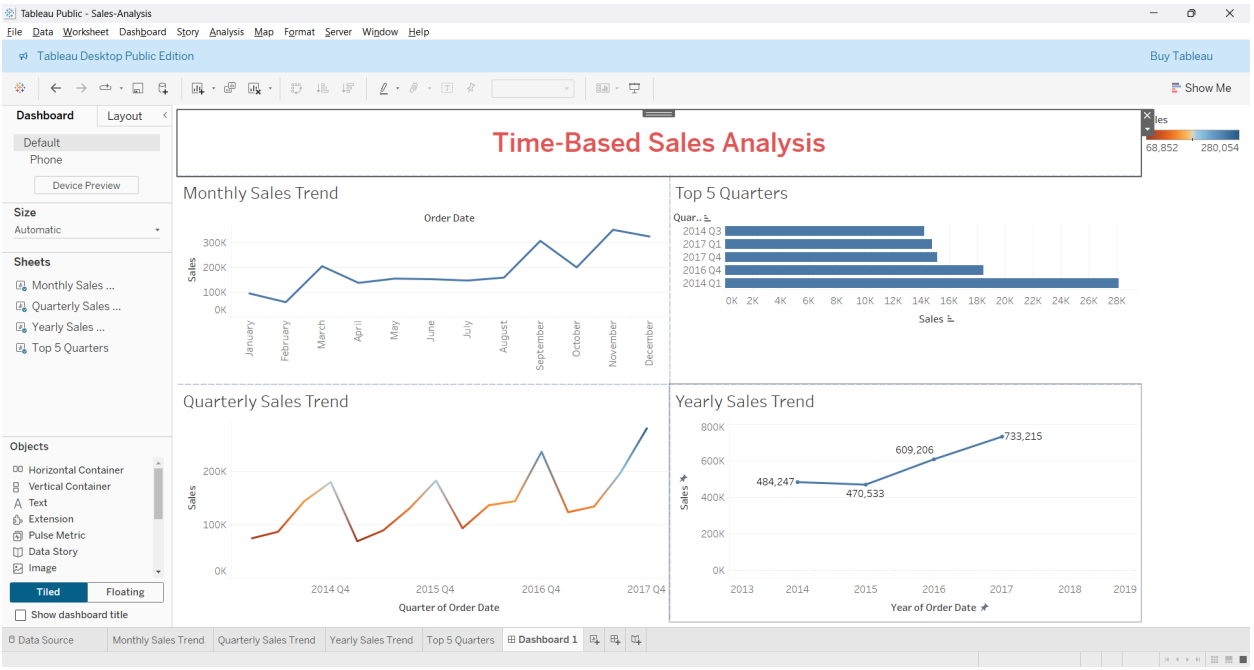
This identifies which specific 3-month periods were the most successful in company history.

1. Click the **New Worksheet** icon.
2. Drag **Order Date** to the **Rows** shelf.
3. Click the arrow on the pill and select the "Quarter"
4. Drag **Sales** to the **column** .
5. **Filter for Top 5:**
    - Drag **Order Date** to the **Filters** shelf. (choose Month /Day /Year)
    - Select **Top** --> **By Field** --> **Top 5** based on **Sales (Sum)**.
6. Sort the results descending by Sales.
7. Rename the sheet: Top 5 Quarters.



Assemble the Time-Based Dashboard

1. Click the **New Dashboard** icon.
2. Set the **Size** to **Automatic**.
3. **Layout Suggestion:**
   - Place the **Monthly Trend** (the most detailed) across the top half.
   - Place the **Quarterly Trend** and **Yearly Trend** side-by-side in the middle.
   - Place the **Top 5 Table** at the very bottom.
4. **Add a Date Filter:**
   - Click on any chart, click the down arrow --> **Filters** --> **Order Date**.
   - Right-click the new filter --> **Apply to Worksheets** --> **All Using This Data Source**.
   - Change the filter style to **"Range of Dates"** or **"Slider"**.
5. **Final Polish:** Add a dashboard title Time-Based Sales Analysis.



d. Profitability Analysis Dashboard

i. Profit by Product Category: A bar chart showing the profit by product category.
ii. Profit by Region: A map showing the profit by region.
iii. Top 5 Products by Profit: A table showing the top 5 products by profit.
iv. Gross Margin by Quarter: A line chart showing the gross margin by quarter.

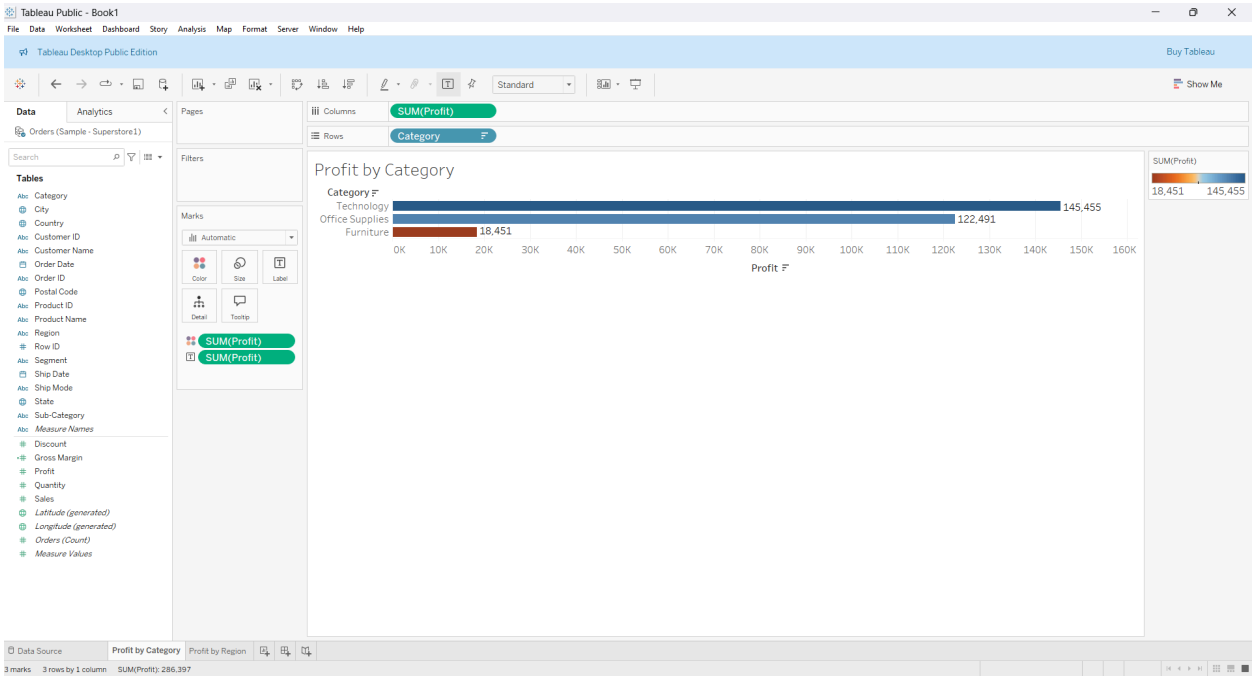
Step 0: Create the "Gross Margin" Calculation

Before building the sheets, you need a measure for Gross Margin.

1. Click the down arrow in the Data Pane and select **Create Calculated Field**.
2. Name it: Gross Margin.
3. Formula: SUM([Profit]) / SUM([Sales])
4. Right-click the new Gross Margin field --> **Default Properties** --> **Number Format** --> **Percentage**.

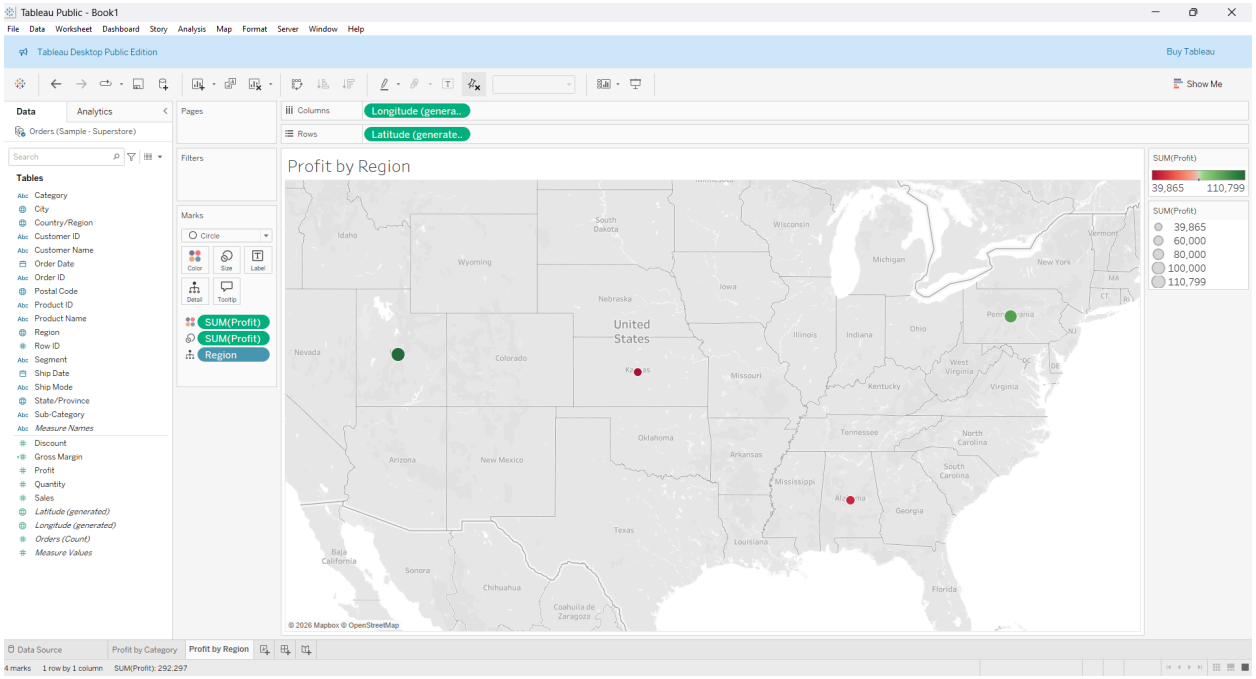
Sheet 1 – Profit by Product Category (Bar Chart)

1. Click the **New Worksheet** icon.
2. Drag **Category** to the **Rows** shelf.
3. Drag **Profit** to the **Columns** shelf.
4. Drag **Profit** to the **Color** card. And to **Label** card
5. Click **Sort Descending** to show the most profitable category at the top.
6. Rename: `Profit by Category`.



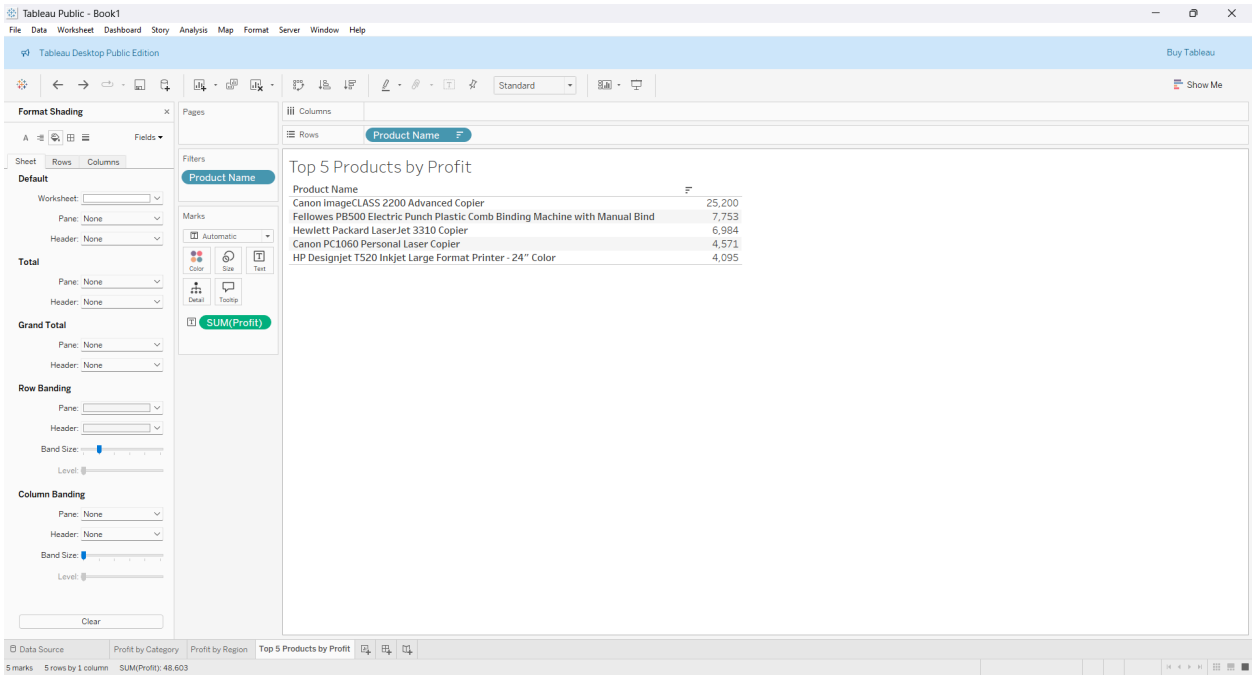
Sheet 2 – Profit by Region (Map)

1. Click the **New Worksheet** icon.
2. **Change Geographic Role:** In the Data Pane (left), right-click **Region** --> **Geographic Role** --> select **State/Province**.
3. **Edit Locations:**
    a. Click the **"4 unknown"** (or 5 unknown) gray pill in the bottom right of your map.
    b. Select **Edit Locations...Country/Region** → United States
    c. In the "Matching Location" column, click the cell next to each region and type a central state:
        i. **Central:** Kansas
        ii. **East:** Pennsylvania
        iii. **South:** Alabama
        iv. **West:** Utah
4. **Visuals:**
    a. Drag **Profit** to **Color** and **Size**.
    b. Change Mark type to **Circle**.
    c. Click **Color** --> **Edit Colors** --> choose **Red-Green Diverging**. (Red = Loss, Green = Profit).



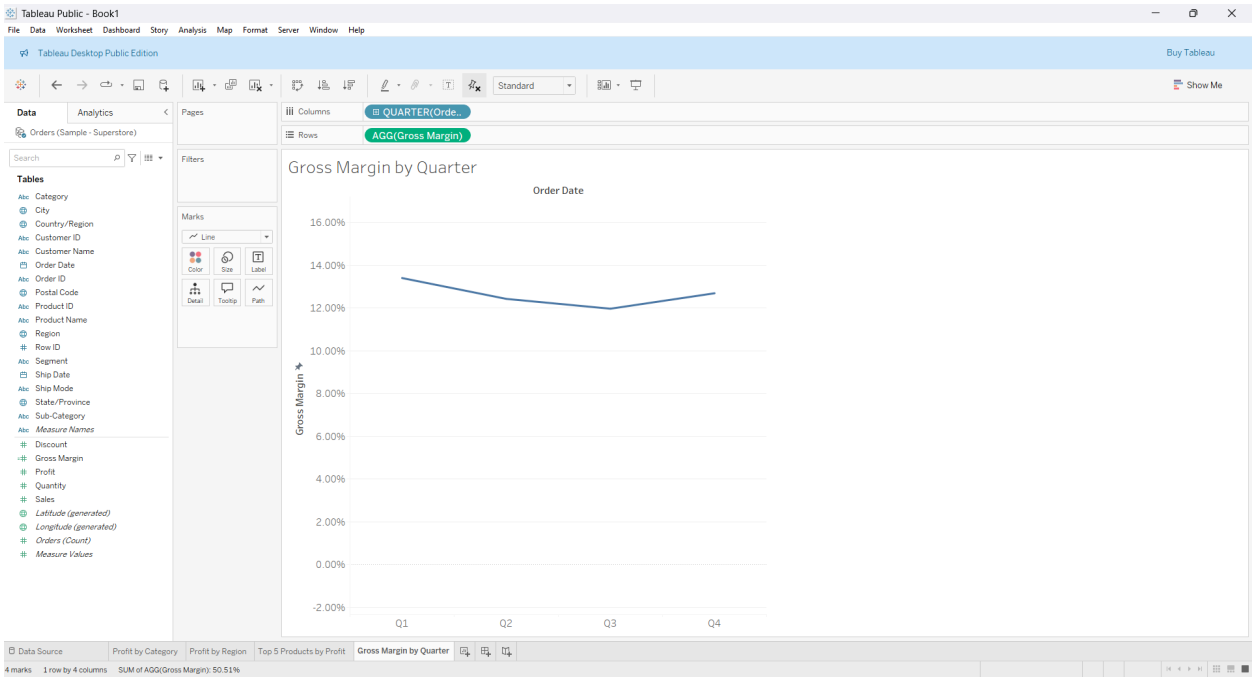
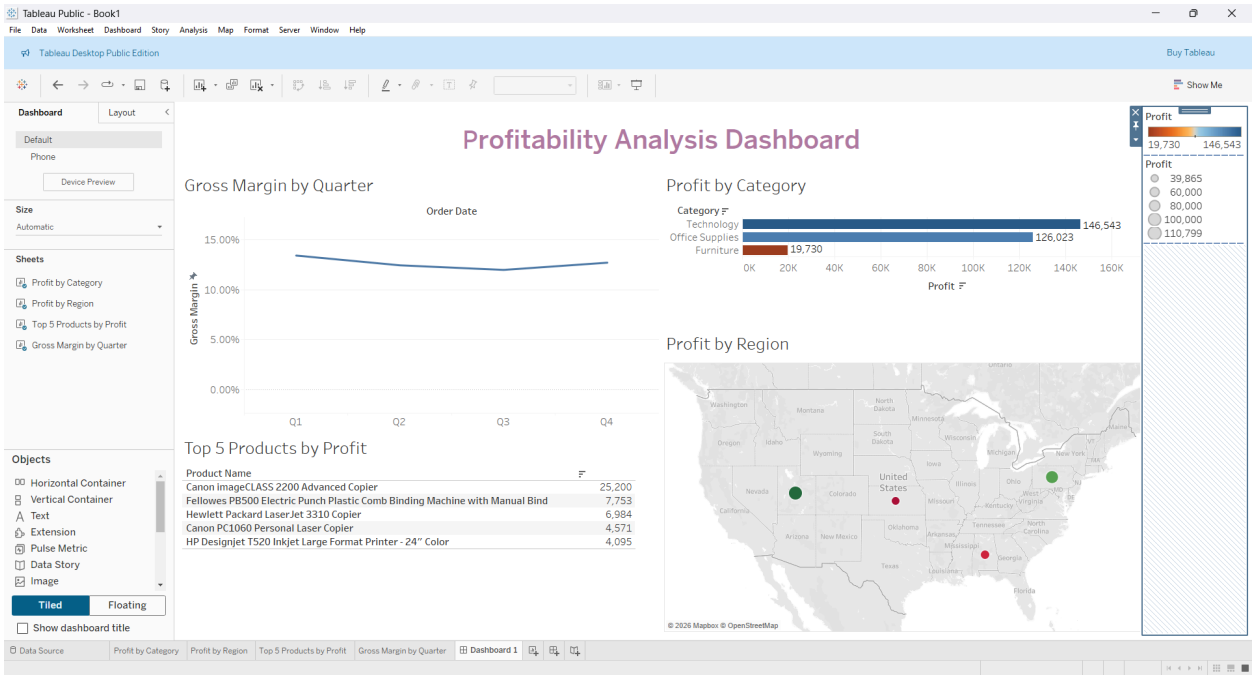
Sheet 3: Top 5 Products by Profit (Table)

1. Open a new worksheet.
2. Drag **Product Name** to the **Rows** shelf.
3. Drag **Profit** to the **Text** card.
4. **The Filter:** * Drag **Product Name** into the **Filters** shelf.
    - Go to the **Top** tab --> select **By field** --> **Top 5** by **Profit (Sum)**.
5. Sort the list by clicking the icon on the **Profit** column header.



Sheet 4: Gross Margin by Quarter (Line Chart)

1. **Build Chart:**
    - Drag **Order Date** to **Columns**. Right-click the pill and select **"Quarter"**
    - Drag **Gross Margin** to **Rows**.
    - Set Mark type to **Line**.



Final Dashboard